PREP 1 · LESSONS 1–2

Inside the atom

Protons, neutrons and electrons · atomic and mass numbers · isotopes and ions

SCORE / 10

Name _____ Class _____ Due _____

Do Now REMEMBER

From earlier lessons · 1 mark each

1 Define electric current. [1]

2 Name the two quantities that kinetic energy depends on. [1]

3 State the frequency and potential difference of the UK mains electricity supply. [1]

Key ideas LESSONS 1–2

Show your working

4 Complete the table. Choose from: **+1** **0** **−1** **1** **very small**

Particle	Relative charge	Relative mass
Proton	+1	1
Neutron	_____	1
Electron	_____	_____

Fill in the three gaps in the table. [2]

5 An atom of sodium is $^{23}_{11}Na$

Number of protons: _____ neutrons: _____ electrons: _____ [2]

6 Define isotopes. [1]

7 Which nucleus is an isotope of $^{12}_{6}C$? | $^{14}_{7}N$ | $^{14}_{6}C$ | $^{12}_{5}B$ | [1]

8 When an atom loses an outer electron, it becomes a | **positive** | **negative** | ion. [1]